

ASGN6

Group 28

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1 Aim

1. To design a J-K master-slave flip-flop using NAND gates and verify its functionality.
2. To convert the J-K flip-flop into a T flip-flop and demonstrate that it divides the input clock frequency by 2.
3. To design a J-K latch using NAND gates and connect an edge detection circuit that generates a narrow pulse at every rising edge of the clock.
4. To verify the correct operation of the edge detection circuit.

2 Apparatus

- IC 74LS10 – Triple 3-input NAND gate
- IC 74LS00 – Quad 2-input NAND gate
- IC 74LS04 – Hex inverter
- Clock / rectangular waveform generator
- Breadboard / digital logic simulator
- Connecting wires
- Power supply
- Oscilloscope / logic analyzer

3 Theory

3.1 J-K Flip-Flop

The J-K flip-flop is a sequential logic circuit derived from the S-R flip-flop. The main advantage of the J-K flip-flop is that it eliminates the invalid state present in the S-R flip-flop.

It has two inputs J and K and two outputs Q and $\bar{Q}$. The next state of the flip-flop depends on the input values and the previous state.

J	K	Q_{next}
0	0	Q_{prev}
0	1	0
1	0	1
1	1	$\bar{Q}_{prev}$

When both inputs are HIGH ($J = K = 1$), the output toggles.

3.2 Race Around Condition

In a level triggered J-K flip-flop, if $J = K = 1$ and the clock pulse remains HIGH for a duration longer than the propagation delay of the circuit, the output toggles repeatedly during the same clock pulse.

This repeated switching of the output during a single clock pulse is known as the **race-around condition**. As a result, the final output state becomes unpredictable.

To eliminate this problem, the **master-slave flip-flop configuration** is used.

3.3 Master-Slave J-K Flip-Flop

A master-slave flip-flop consists of two latches connected in series:

- Master latch
- Slave latch

These two latches operate on opposite phases of the clock.

When the clock is HIGH

- The master latch becomes active and captures the input values.
- The slave latch remains disabled.

When the clock transitions from HIGH to LOW

- The master latch becomes disabled.
- The slave latch becomes active.

- The stored value is transferred to the output.

Therefore the output changes only once during each clock cycle, eliminating the race-around condition.

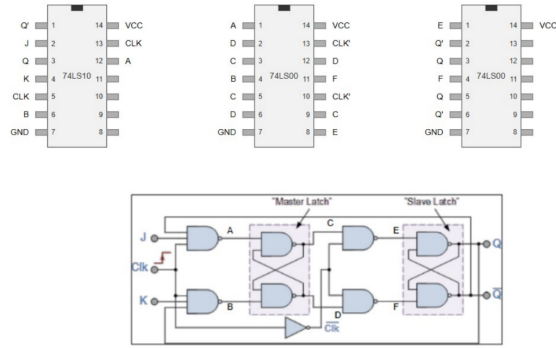


Figure 1: J-K Master Slave Flip-Flop using NAND gates

3.4 Propagation Delay

In digital circuits, every logic gate requires a small amount of time for the output to respond to a change in the input. This delay is known as **propagation delay** and is represented as t_{pd} .

If a signal passes through multiple gates, the total delay becomes approximately the sum of the individual delays.

$$t_{total} \approx n \times t_{pd}$$

where n is the number of gates in the signal path.

This delay ensures that the master latch stores the input before the slave latch updates the output.

3.5 Conversion of J-K Flip-Flop to T Flip-Flop

A T flip-flop is a single input flip-flop that toggles its output when the input is HIGH.

A J-K flip-flop can be converted into a T flip-flop by connecting

$$J = K = T$$

The operation becomes:

T	Q_{next}
0	Q_{prev}
1	$\overline{Q_{prev}}$

Thus when $T = 1$, the output toggles at every clock pulse.

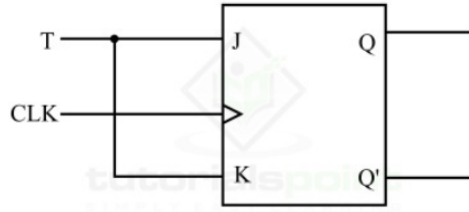


Figure 2: Conversion of J-K Flip-Flop to T Flip-Flop

3.6 Frequency Division

Consider a clock signal with frequency f_{in} applied to the T flip-flop.

Each clock pulse toggles the output once. However, one complete output cycle requires two toggles:

- One transition from 0 to 1
- One transition from 1 to 0

Therefore one complete output cycle occurs after two input clock cycles. Hence the output frequency becomes

$$f_{out} = \frac{f_{in}}{2}$$

Thus the T flip-flop acts as a **divide-by-2 frequency divider**.

3.7 Edge Detection Circuit

A latch normally responds whenever the enable signal is active. To ensure that the latch responds only during the rising edge of the clock, an edge detection circuit is used.

The clock signal is sent through two paths:

- One path carries the original clock signal.
- The second path carries the clock signal through an inverter, which introduces a small propagation delay.

For a short interval after the clock rises, the original clock is HIGH while the delayed signal is still LOW. This creates a narrow pulse at the output of the logic gate.

This pulse is used as the enable signal for the latch so that the latch changes state only at the rising edge of the clock.

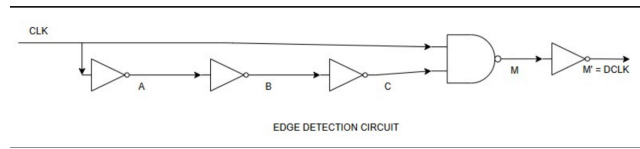


Figure 3: Edge Detection Circuit

4 Procedure

1. Construct the J-K master-slave flip-flop using NAND gates.
2. Apply different combinations of inputs J and K.
3. Provide a clock signal and observe outputs Q and $\overline{Q}$.
4. Verify the truth table of the J-K flip-flop.
5. Connect $J = K = T$ to convert the circuit into a T flip-flop.
6. Apply a rectangular clock waveform.
7. Observe the output waveform and verify that the output frequency is half the input frequency.
8. Construct the J-K latch using NAND gates.
9. Implement the edge detection circuit using inverter and logic gates.
10. Connect the generated pulse to the enable input of the latch.
11. Verify that the latch changes state only at the rising edge.

5 Observation

5.1 J-K Flip-Flop

J	K	Q_{next}
0	0	Q_{prev}
0	1	0
1	0	1
1	1	Toggle

5.2 T Flip-Flop

T	Q_{next}
0	Q_{prev}
1	Toggle

Observed waveform verifies that

$$f_{out} = \frac{f_{in}}{2}$$

6 IC Connection Diagram

The IC pin configuration used in the experiment is shown below.

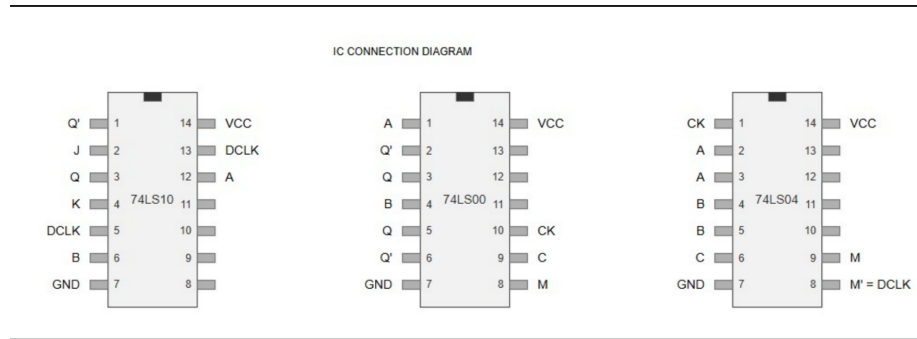


Figure 4: IC connection diagram for 74LS10, 74LS00 and 74LS04

7 Result

- The J-K master-slave flip-flop using NAND gates was successfully implemented.
- The race-around condition was eliminated using the master-slave configuration.
- The J-K flip-flop was converted into a T flip-flop by connecting $J = K = T$.
- The output waveform confirmed that the circuit divides the input frequency by 2.
- The edge detection circuit successfully generated a narrow pulse at each rising clock edge.